

CITY: LONDON

EMAIL: KASADLUGOS@GMAIL.COM

TEL" +44 7927 810 613

Katarzyna Długosz

Design engineer working across soft robotics, physical computing, smart textiles and computer vision. Background in technical garment design and B2B product development; completing an MA at UAL (Nov 2026). I specialise in bridging physical qualities of fabric with soft robotics, wearable technology and computer vision.

SKILLS

CAD SOFTWARES:

Adobe Creative Suite | Affinity | Fusion 360 | Blender | Clo 3D | Unity

OTHER SOFTWARE:

VS Code | Arduino IDE | EasyEDA | MuJoCo | Nvidia Isaac Lab | Streamlit | Pydantic | LLM APIs

HARD SKILLS:

Engineering & ML:

Python, forward and inverse kinematics, humanoid pose estimation, vision models, basic C++ and C#

Physical Computing:

Smart textiles, sensor integrations, robotic prototyping and pneumatic systems

Design & Fabrication:

3D modelling, simulation and rendering, 3D printing and silicone casting

Garment Engineering:

Digital and physical patterncutting, technical design, developing grading systems

Relations & Production Management:

B2B communication and tracking, supplier sourcing and relationship management,

EDUCATION

Design & Technology - Master of Arts

University of the Arts, London

2025- Nov 2026 - ongoing

Bachelor of Arts

University of the Arts, London College of Fashion, London

2022 - 2025, First Class Honours

ACHIEVEMENTS

2026 | Robotic Craftsman was featured in Microsoft x UAL documentary

2025 | CFG Coding Python Certificate

2025/6 | Monsoon Accessorize Scholarship for circular design

2025/6 | The Leathersellers Foundation Power2Aspire programme - mentoring award for sustainable design practitioner

2024 | LCF x Marks & Spencer Industry Challenge - winner

2023 | NEOFUR FRANCE sustainable design competition - finalist

WORK EXPERIENCE

2025 - now | Technical Designer, B Fashion Studio

Responsible for end-to-end technical design, product development and raw materials sourcing for independent clients. Leading day-to-day conversations with multiple B2B clients and suppliers. Built website and correspondence add-ins.

2025 | Atelier Assistant, Robert Wun

Assisting the Atelier Team with pattern amendments and making of the 2026 Haute Couture Collection

2024 | Design Consultant /Intern, CYA

Responsible for product development from pattern stage to large-scale manufacturing for an innovative lingerie start-up brand, including developing a new sizing system, market research and sourcing materials

2024 | Design and Production Intern, Galia Lahav

Assisting the RTW team with technical design, documentation and tracking production supply chain

LANGUAGES

POLISH - Native | ENGLISH - Highly Fluent, IELTS Academic Certificate C2 | RUSSIAN - Basic

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PROJECTS

Hania – The Robotic Craftsman (2026) · <https://github.com/kat2137/robotic-craftsman>

Robotic arm and tendon-driven hand built to learn hand-sewing, asking what gets lost when a craft skill becomes data. 17 joints across arm and hand; 6 joint systems prototyped; 20 makers filmed sewing as training data. Hand-pose estimation (WiLoR) mapped onto a MuJoCo simulation, with kinematics built from Fusion 360 CAD. Python, Arduino, LeRobot/SmolVLA, NVIDIA Isaac Lab, Jetson Orin Nano. Supported by the RS UK Student Fund; featured in a Microsoft x UAL documentary.

Tech pack generator (2026) · https://github.com/kat2137/tech_pack_auto

Tool that turns garment drawings into stitch breakdowns, seam types and bills of materials, restricted to a fixed set of standard stitch types so the output matches what factories use. Used for 10–20 tech packs a week. Python, Streamlit, Pydantic, Anthropic API.

PneumaBra (2026) · <https://github.com/kat2137/pneumabra>

Bra whose support the wearer adjusts with air to follow changes across the hormonal cycle, the working day and activity. Silicone air channels cast into the softest candidate fabric, chosen by tensile testing. Firmware (Arduino C++, ~660 lines) on an ESP8266 serves a built-in web app: it tracks the cycle day, sets a target pressure for each day, and drives the pump from pressure-sensor readings across 3 modes. Developed with 4 users studied in depth; 2 mm channel creep over 10 hours. Fusion 360, PlatSil silicone, EasyEDA PCB.

Sviatovid (2025) · https://github.com/kat2137/Googly_eyes_project

Wearable leather bag - "head" whose eyes find and follow onlookers, after the all-seeing Slavic deity. Real-time face tracking on a Raspberry Pi 4 (OpenCV detection, with the eyes re-centred on the nearest face), driving a two-axis pan-and-tilt eye mechanism through a PCA9685 servo driver controlled directly over I²C. Camera, Pi, driver and battery packed into a 151 × 51 mm housing, built to be serviceable. Python, OpenCV, Fusion 360.

Sony Interactive Entertainment – MA Collaborative Challenge (2025) · https://github.com/kat2137/group_b_project

Location-based game for Sony's modular accessibility controller. Wrote the Unity/C# control layer (~1,000 lines) driving the controller's lights, sound and haptic modules across 6 game modes (compass navigation, torch, emergency, social). Group project.